

GOMEL, BelarusWMO: **330410**Lat: **52.4019N**Lon: **30.9631E**Elev: **414**StdP: **14.48**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.6	1.2	-9.4	3.4	-4.0	-3.9	4.5	1.7	23.6	30.1	21.3	31.1	5.8	90	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	87.7	68.5	84.3	67.4	81.3	65.8	70.8	82.4	69.3	80.3	67.7	78.1	8.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	101.8	75.0	65.8	96.6	73.6	64.2	91.3	72.0	34.9	82.6	33.6	80.2	32.4	78.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.5	17.4	15.7	DB	-11.1	92.0	6.3	3.5	-15.6	94.5	-19.3	96.5	-22.9	98.5	-27.5	101.0
(5)				WB	-11.4	73.9	6.3	2.2	-15.9	75.5	-19.6	76.8	-23.1	78.1	-27.7	79.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	23.2	24.9	33.8	48.2	58.7	64.8	68.6	66.5	56.4	45.2	35.3	26.7
	DBStd	18.22	10.72	10.81	8.39	7.81	8.03	6.57	5.54	6.29	7.11	7.62	8.80	10.30
	HDD50	3552	830	702	503	125	18	0	0	21	189	442	722	
	HDD65	7244	1295	1122	966	506	226	85	27	55	270	615	890	1187
	CDD50	2146	0	0	2	70	289	444	575	512	213	39	2	0
	CDD65	363	0	0	0	1	32	78	138	102	12	0	0	0
	CDH74	3335	0	0	0	13	303	692	1220	984	122	1	0	0
	CDH80	957	0	0	0	1	59	169	373	338	17	0	0	0
Wind	WSAvg	7.9	8.6	8.4	8.6	7.9	7.4	7.4	6.8	6.9	7.2	7.9	8.2	8.8
Precipitation	PrecAvg	25.2	1.5	1.4	1.5	1.6	2.5	3.0	3.7	2.2	2.2	2.2	1.9	1.6
	PrecMax	34.7	2.7	2.7	3.3	4.3	5.5	5.2	8.7	6.4	4.8	5.4	4.3	3.4
	PrecMin	20.0	0.5	0.5	0.3	0.1	1.4	1.2	1.3	0.3	0.1	0.2	0.1	0.4
	PrecStd	3.5	0.6	0.5	0.7	1.1	0.9	1.1	1.8	1.3	1.2	1.4	1.0	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	46.3	60.6	75.2	85.7	88.0	91.6	94.8	82.3	71.3	55.7	48.5
		MCWB	41.8	41.9	47.9	56.8	65.0	68.2	70.1	69.0	63.5	58.5	52.6	46.9
	2%	DB	40.0	41.6	55.0	70.2	81.0	84.5	87.6	87.8	77.4	64.4	51.9	44.5
		MCWB	38.5	39.0	45.3	54.6	63.8	66.7	69.0	68.5	62.4	56.7	49.0	43.3
	5%	DB	37.5	39.0	49.7	66.0	77.2	81.4	84.2	83.5	73.2	60.0	48.7	41.2
		MCWB	36.1	37.2	42.6	53.1	62.3	65.5	68.0	67.2	60.7	54.2	46.7	40.4
	10%	DB	35.3	36.5	45.0	61.6	73.4	78.4	80.8	79.1	68.8	56.5	46.3	37.7
		MCWB	34.3	34.9	40.3	50.8	60.1	64.1	66.9	64.8	58.7	52.0	44.8	36.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.1	42.9	49.7	59.9	68.7	71.0	73.5	72.5	66.6	61.2	53.4	47.3
		MCDB	43.2	45.2	59.2	71.0	80.9	82.2	84.6	85.0	77.4	67.1	55.7	48.0
	2%	WB	38.5	39.4	45.9	56.5	65.7	68.8	71.3	70.2	63.8	57.6	50.0	43.0
		MCDB	39.4	41.2	52.3	66.7	77.5	80.2	82.8	82.6	74.0	62.7	51.4	43.9
	5%	WB	36.1	37.0	43.2	54.3	63.5	67.2	69.8	68.6	61.7	55.0	47.0	40.2
		MCDB	37.1	38.3	48.0	63.4	74.1	77.9	80.2	80.5	70.4	59.4	48.3	41.0
	10%	WB	34.0	35.1	40.6	52.1	61.5	65.5	68.2	66.8	59.7	52.5	44.6	36.7
		MCDB	34.8	36.2	44.6	60.3	71.2	74.9	77.8	76.8	66.8	56.2	45.7	37.5
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.7	13.6	17.5	19.3	18.9	18.8	19.3	17.6	13.7	8.4	8.0
		MCDBR	7.1	11.0	20.6	23.7	23.6	23.3	23.8	25.0	23.7	19.1	11.0	8.3
		MCWBR	6.6	9.0	12.7	11.5	10.1	9.6	9.2	9.7	11.0	12.0	8.7	7.4
	5% WB	MCDBR	6.8	9.7	17.8	21.3	22.4	20.6	20.9	22.0	21.0	16.6	9.5	8.2
MCWBR		6.6	8.2	12.1	11.2	10.4	9.8	9.2	9.8	10.8	11.4	8.0	7.5	
Clear-Sky Solar Irradiance	taub	0.302	0.336	0.367	0.427	0.405	0.392	0.423	0.429	0.390	0.360	0.340	0.305	
	taud	2.147	2.085	2.148	2.137	2.281	2.340	2.269	2.236	2.308	2.360	2.261	2.186	
	Ebn at Noon	210	237	257	255	269	273	261	252	248	231	196	186	
	Edh at Noon	23	33	38	43	40	38	40	39	32	25	21	19	
All-Sky Solar Radiation	RadAvg	247	513	961	1333	1744	1846	1783	1523	1038	545	207	155	
	RadStd	50	93	136	134	120	154	152	119	113	102	36	39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.17	N/A	N/A	+1.78	+1.04	+0.92	-227	-380	+193	+85
Regional (8 neighbors)	+1.13	N/A	N/A	N/A	+1.08	N/A	-220	-349	+180	+65